

Injuries related to electric scooter and bicycle use in a Washington, DC, emergency department

Jessica B. Cicchino, Paige E. Kulie & Melissa L. McCarthy

To cite this article: Jessica B. Cicchino, Paige E. Kulie & Melissa L. McCarthy (2021) Injuries related to electric scooter and bicycle use in a Washington, DC, emergency department, Traffic Injury Prevention, 22:5, 401-406, DOI: [10.1080/15389588.2021.1913280](https://doi.org/10.1080/15389588.2021.1913280)

To link to this article: <https://doi.org/10.1080/15389588.2021.1913280>

 View supplementary material 

 Published online: 07 May 2021.

 Submit your article to this journal 

 Article views: 36

 View related articles 

 View Crossmark data 



Injuries related to electric scooter and bicycle use in a Washington, DC, emergency department

Jessica B. Cicchino^a , Paige E. Kulie^b, and Melissa L. McCarthy^c

^aInsurance Institute for Highway Safety, Arlington, Virginia; ^bDepartment of Emergency Medicine, George Washington University Medical Center, Washington, DC; ^cMilken Institute School of Public Health, George Washington University, Washington, DC

ABSTRACT

Objective: E-scooter use has grown rapidly in the United States. Its rise in popularity has coincided with the promotion of cycling in many cities, but more needs to be known about how these transportation modes compare to determine if cycling should serve as an appropriate benchmark for policy decisions and safety expectations regarding e-scooters.

Methods: We examined characteristics of adults seeking treatment in a Washington, DC, emergency department (ED) for injuries associated with riding e-scooters during 2019 ($n = 99$) or bicycles during 2015–2017 ($n = 337$).

Results: E-scooter incidents less frequently involved moving vehicles (13.1% vs. 37.7%) or occurred on roads (24.5% vs. 50.7%) than cycling incidents. A smaller proportion of injured e-scooter riders were ages 30–49 (32.3% vs. 48.4%) and a larger proportion were 50 and older (34.3% vs. 22.6%) or female (45.5% vs. 29.1%). Distal lower extremity injuries were more common among e-scooter riders (13.1% vs. 3.0%; RR, 2.76; 95% CI, 1.79–3.54), and injuries to the proximal upper extremity (9.1% vs. 20.5%; RR, 0.49; 95% CI, 0.24–0.92) or chest, abdomen, and spine (3.0% vs. 14.0%; RR, 0.24; 95% CI, 0.07–0.70) were less common. Head injury rates were similar, but e-scooter riders more often experienced concussion with loss of consciousness (4.0% vs. 0.6%; RR, 3.03; 95% CI, 1.20–4.09) and were far less likely to wear helmets (2.0% vs. 66.4%). Estimated ED presentation rates per million miles traveled citywide were higher among e-scooter riders than cyclists (RR, 3.76; 95% CI, 3.08–4.59).

Conclusions: E-scooters and bicycles are both popular forms of micromobility, but the characteristics of riders injured on them, the ways in which they become injured, and the types of injuries they sustain differ substantially. E-scooter rider injury rates, though currently high, may decrease as they gain experience; however, if the number of new users continues to climb, they will persist in using the ED more often than cyclists per mile that they travel.

ARTICLE HISTORY

Received 29 September 2020

Accepted 31 March 2021

KEYWORDS

Micromobility; e-scooter; injury; bicycle; shared mobility

Introduction

Shared micromobility has grown rapidly in the United States, with 38.5 million trips logged on shared standing electric scooters (e-scooters) in 2018 and 86 million trips in 2019 (National Association of City Transportation Officials 2020). An estimated 14,000 e-scooter riders were treated in U.S. emergency departments (EDs) and over 1,000 were hospitalized in 2018 (Namiri et al. 2020).

Cities have grappled with how to fold e-scooters into existing infrastructure and regulation as this new mode has proliferated. Bicycles arguably represent a similar form of transportation, given their use within shared micromobility programs and suitability to urban transport. Accordingly, some cities subject e-scooter riders to the same rules as cyclists concerning behavior such as sidewalk riding or encourage e-scooters to use bike lanes.

Evidence comparing the circumstances leading to injuries, injury rates, and injury types of cyclists and e-scooter riders is needed to assess if cycling should serve as an appropriate benchmark for policy decisions and safety expectations regarding e-scooters. Early research has demonstrated some differences

between them. E-scooter riders treated in EDs nationally during 2000–2017 were more likely to sustain fractures than cyclists (DiMaggio et al. 2020). During 2017–2018, Watson et al. (2020) reported differences in the distribution of body parts injured by e-scooter and cyclists presenting to EDs in a national sample and estimated higher injury rates per hour traveled among e-scooter riders than cyclists.

The goal of the current study was to compare personal and injury characteristics between e-scooter riders and cyclists presenting to a single ED in Washington, DC, and to assess estimated rates of presentation to the ED per mile traveled citywide by these modes. By November 2019, eight operators deployed over 4,600 e-scooters in Washington, DC, and the city was among the top six in ridership in the United States (National Association of City Transportation Officials 2020). While differences between injured cyclists and e-scooter riders have been identified nationally, it is unclear how they would persist within a single area where riders of both modes are exposed to similar environmental characteristics; e-scooters are available in select cities, while national statistics for cycling injuries include those occurring in urban, suburban,

and rural areas. Additionally, while emerging research has begun to identify the types of injuries sustained by e-scooter riders treated in emergency departments (e.g., Badeau et al. 2019; Trivedi et al. 2019; Puzio et al. 2020), most has consisted of retrospective chart reviews and less is known about the circumstances under which e-scooter riders are injured. We combined rider interviews with medical record reviews to provide a richer picture of how these injuries occur.

Methods

Data collection

Shared e-scooters were first deployed in Washington, DC, in March 2018. We prospectively recruited injured e-scooter riders March 23 to November 30, 2019 who sought treatment at the ED of George Washington University Hospital, an urban academic medical center with an annual ED census of 75,000 patient visits. We had previously recruited cyclists April 14, 2015 to September 30, 2017 presenting to the same ED as part of an earlier multisite study on infrastructure (Cicchino et al. 2020), and we compare the e-scooters recruited in 2019 to this group of cyclists in this paper. To match groups on seasonality, the cyclist sample was limited to those injured from March to November. The George Washington University Institutional Review Board approved both protocols.

E-scooter riders and cyclists were required to be age 18 years or older and English-speaking to be eligible to enroll, and were ineligible if they were unable to communicate with ED staff or understand consent, injured outside the hospital's catchment area, injured more than one week prior, left the ED without being seen by a physician, were previously enrolled for the same trip, or were riding with more than one person on the same bicycle or e-scooter (Figure A1, online supplement). Cyclists were additionally excluded if they were trick riding or racing. We also collected data from nonriders injured by e-scooters (e.g., pedestrians that tripped over scooters) not reported here. E-scooter riders were compensated \$10 for participating; cyclists were not compensated. Participants were screened and enrolled in the ED during hours research assistants were present, which were 8:00 AM (during 2015–2017) or 9:00 AM (during 2019) to 10:00 PM, or in the hospital if admitted. The original goal of collecting data on e-scooter riders was to examine their characteristics independently, which is why the eligibility criteria and compensation did not exactly match that for cyclists. Although the differences in time periods of data collection and eligibility criteria were not ideal, they allowed for a cost-effective opportunity to compare these groups of road users by drawing on existing data on cyclists.

Trained research assistants interviewed enrolled participants using a structured questionnaire that asked about demographics, riding history, trip purpose, and circumstances and geographic location of their injury. The questionnaire for e-scooter riders was derived from the survey previously administered to cyclists. Participants were asked to trace the route leading to their injury on an electronic map, from which trip length was derived. Incident circumstances were determined from an open-ended question asking participants to describe how they were injured and a series of forced-choice questions asking if other road users or environmental characteristics contributed to their crash or fall.

If a participant was injured at an intersection, the location of their incident was considered to be where they were riding prior to reaching the intersection. A research assistant abstracted medical records of participants and coded each injury sustained using the Abbreviated Injury Scale (AIS) (Gennarelli and Wodzin 2008). AIS coding was checked for errors by a second researcher.

Estimated ED visits per mile traveled

We estimated rates of presentation to the ED per mile traveled using all screened adults who were injured while riding an e-scooter or bicycle in the hospital's catchment area in the past week during March to November in the numerator (Figures A2 and A3, online supplement).

We obtained the number of trips and miles ridden by e-scooters in Washington, DC, during the study period from the District Department of Transportation (DDOT), which received monthly trip reports from shared e-scooter operators. The number of bicycle trips citywide was extrapolated from the average number of bicycle commuters in Washington, DC, during 2015 to 2017 as reported by the American Community Survey (United States Census Bureau 2015, 2016, 2017). We multiplied the number of e-scooter and cycling trips estimated citywide by the mean trip lengths prior to injury among study participants to determine total miles traveled. The appendix in the online supplement describes our methods for estimating the number of trips and calculating rates in greater detail. We performed sensitivity analyses, also described in the appendix in the online supplement, that varied assumptions to extrapolate the number of bicycle trips from commuting data and restricted the sample to exclude riders who did not complete their entire trip within Washington, DC.

Analysis

We compared enrolled e-scooter riders and cyclists by trip and demographic characteristics using the χ^2 test for proportions and the t test for means. Logistic regression was used to compare riders on evaluation, treatment, and injury characteristics, controlling for age and gender. Odds ratios from logistic regression models were transformed to relative risks using the method of Zhang and Yu (1998). We also present rate ratios and corresponding 95% confidence intervals (CIs) comparing estimated ED presentation rates per mile traveled between groups.

Results

Demographic and trip characteristics

During the 2015–2017 study period, 561 individuals presented to the ED who were potentially injured while cycling, and during the 2019 period, 148 individuals presented who were potentially injured by e-scooter (Figure A1, see online supplement). We excluded 32 (7 possible e-scooters, 25 possible cyclists) who were missed prior to screening, 151 (31 possible e-scooters, 120 possible cyclists) who were ineligible, and 90 who were eligible but refused to participate or withdrew (11 e-scooters, 79 cyclists). The sample of 436 interviewed participants included 99 e-scooter riders and 337 cyclists.

Compared with cyclists, interviewed e-scooter riders were more often female, less often ages 30–49, and more often age 50 or older (Table 1). A greater proportion of their trips occurred on weekends or at night. E-scooter riders less often wore helmets (2 e-scooter riders [2.0%] vs. 223 cyclists [66.4%]; $P < .001$). Their trips were more frequently for social purposes (35 [35.4%] vs. 36 [10.7%]; $P < .001$), while the majority of cyclists (175 [52.1%]) were commuting (Table 2). E-scooter riders less frequently were on the road prior to their injury (24 [24.5%] vs. 171 [50.7%]; $P < .001$) and most often were on the sidewalk (55 [56.1%]). Far fewer e-scooter riders than cyclists were injured in incidents involving moving vehicles (13 [13.1%] vs. 127 [39.5%]; $P < .001$); e-scooter riders were most commonly injured because of hazardous surface features (24 [24.2%]), infrastructure such as curbs (16 [16.2%]), or other falls (26 [26.3%]) (Table 3). See Table A1 in the online supplement for an expanded comparison of the demographic and trip characteristics of e-scooter riders and cyclists.

E-scooter trips prior to injury were on average about one-fourth the length of cycling trips (mean [SD] miles, 1.05 [1.13] vs. 4.10 [6.02]; $P < .001$). About a third (35 [35.7%]) of e-scooter riders were using a scooter for the first time, while most (265 [78.9%]) cyclists reported biking most days of the week during the season they predominantly ride.

ED evaluation, treatment, and injury characteristics

Table 4 summarizes ED evaluation, injury characteristics, and treatments performed, and Table A2 (online supplement) describes them in more detail. E-scooter riders were less likely

Table 1. Personal and trip characteristics of cyclists (2015–2017) and e-scooter riders (2019) seeking treatment in a Washington, DC, emergency department [No. (%)].

Characteristic	E-scooter riders (n = 99)	Cyclists (n = 337)	P-value ^a
Age (years)			
18–29	33 (33.3)	98 (29.1)	.01
30–49	32 (32.3)	163 (48.4)	
50+	34 (34.3)	76 (22.6)	
Female	45 (45.5)	98 (29.1)	.002
Wore helmet ^b	2 (2.0)	223 (66.4)	<.001
Weekend	32 (32.3)	58 (17.2)	.001
Time of day ^b			.006
7 AM–2:59 PM	56 (57.1)	232 (68.8)	
3 PM–10:59 PM	30 (30.6)	91 (27.0)	
11 PM–6:59 AM	12 (12.2)	14 (4.2)	

^aDetermined through χ^2 test or proportions.

^bProportions exclude cases with missing data; <2% missing.

than cyclists to arrive by ambulance (39 [39.4%] vs. 176 [52.2%]; relative risk [RR], 0.62; 95% CI, 0.41–0.91). They also less often underwent imaging to the chest, abdomen, or spine (17 [17.2%] vs. 138 [41.0%]; RR, 0.35; 95% CI, 0.20–0.59), and were less likely to have been injured in those areas (3 [3.0%] vs. 47 [14.0%]; RR, 0.24; 95% CI, 0.07–0.70). Sixteen cyclists [4.7%] were diagnosed with rib fractures and 8 [2.4%] with major intra-abdominal or intrathoracic injuries, including 5 with pulmonary contusions, 4 with hemo/pneumothorax, and 3 with liver or spleen lacerations; no e-scooter riders received these diagnoses.

Similar proportions of e-scooter riders and cyclists were diagnosed with head or neck injuries (7 [7.1%] vs. 25 [7.4%]), but the types of head injuries differed. E-scooter riders more often sustained concussions with loss of consciousness (4 [4.0%] vs. 2 [0.6%]; RR, 3.03; 95% CI, 1.20–4.09). Additionally, 2 e-scooter riders and no cyclists experienced skull fractures.

Both groups frequently experienced extremity injuries, with the most common injury to the distal upper extremity. E-scooter riders less often injured the proximal upper extremity than cyclists (9 [9.1%] vs. 69 [20.5%]; RR, 0.49; 95% CI, 0.24–0.92), and more often experienced a distal lower extremity injury (13 [13.1%] vs. 10 [3.0%]; RR, 2.76; 95% CI, 1.79–3.54). A larger proportion of e-scooter riders than cyclists required splinting (47 [47.5%] vs. 109 [32.3%]; RR, 1.69; 95% CI, 1.23–2.20) or fracture or dislocation reduction (12 [12.1%] vs. 15 [4.5%]; RR, 2.06; 95% CI, 1.22–2.94).

Most riders were discharged home from the ED. Of the 9 (9.1%) e-scooter riders and 27 (8.0%) cyclists who were admitted or transferred, 4 cyclists (1.2%) and no e-scooter riders were admitted to the intensive care unit. Similar proportions of e-scooter riders and cyclists sustained no injuries or had maximum AIS scores of 1 (60 [60.6%] vs. 194 [58.3%]) or 2 (37 [37.4%] vs. 122 [36.6%]). Although slightly more cyclists than e-scooter riders had maximum AIS scores of 3 or greater (2 [2.0%] vs. 17 [5.1%]), indicating serious injury, mean Injury Severity Scores (ISS) did not differ significantly (2.6 vs. 3.4; RR, 0.94; 95% CI, 0.88–1.08).

Evaluation, treatment, and injury characteristics were also compared separately between e-scooter riders and cyclists in common injury circumstances for each type of rider (Table 4; see Tables A3 and A4 in the online supplement for more detail). Circumstances examined included incidents involving vehicles (parked or moving) and those due to surface features or infrastructure. Extremity injuries were less prevalent among both rider types in incidents involving vehicles and more frequent in those involving surface conditions or infrastructure. Differences

Table 2. Trip purposes and incident locations of cyclists (2015–2017) and e-scooter riders (2019) seeking treatment in a Washington, DC, emergency department [No. (%)].

Characteristic	E-scooter riders (n = 99)	Cyclists (n = 337)	P-value ^a
Trip purpose ^b			
Social reasons (e.g., movies, visit friends)	35 (35.4)	36 (10.7)	<.001
Commute to/from work	25 (25.3)	175 (52.1)	
Recreation	19 (19.2)	69 (20.5)	
Personal business (e.g., errands)	18 (18.2)	35 (10.4)	
To/from school	0	6 (1.8)	
Other	2 (2.0)	15 (4.5)	
Location ^b			
Sidewalk	55 (56.1)	51 (15.1)	<.001
Road	24 (24.5)	171 (50.7)	
Off-road/multiuse trail	9 (9.2)	51 (15.1)	
Bike lane	8 (8.2)	60 (17.8)	
Alley, driveway, parking lot	2 (2.0)	4 (1.2)	

^aDetermined through χ^2 test or proportions.

^bProportions exclude cases with missing data; <2% missing.

Table 3. Circumstances leading to injury for cyclists (2015–2017) and e-scooter riders (2019) seeking treatment in a Washington, DC, emergency department [No. (%)].

Characteristic	E-scooter riders (n = 99)	Cyclists (n = 337)	P-value ^a
Moving vehicle ^b	13 (13.1)	127 (37.7)	<.001
Parked or stopped vehicle	2 (2.0)	43 (12.8)	
Pedestrian	11 (11.1)	17 (5.0)	
Cyclist or scooter ^c	3 (3.0)	22 (6.5)	
Surface feature (e.g., potholes, uneven pavement)	24 (24.2)	48 (14.2)	
Infrastructure (e.g., curb, pole, fence)	16 (16.2)	18 (5.3)	
Other fall	26 (26.3)	59 (17.5)	
Other circumstance	4 (4.0)	3 (0.9)	

Note: Unless specifically labeled as a fall, circumstances included crashes with and falls to avoid listed element.

^aDetermined through χ^2 test or proportions.

^b10 (10.1%) e-scooter riders and 111 cyclists (32.9%) crashed with moving vehicles.

^c1 e-scooter rider collided with another scooter. Other crashes and falls in this category involved collisions with or falls to avoid cyclists.

Table 4. Evaluation, treatment, and injury characteristics of cyclists (2015–2017) and e-scooter riders (2019) seeking treatment in a Washington, DC, emergency department [No. (%) or Mean $\pm$ SD].

Characteristic	Total		Incident involved motor vehicle		Incident involved infrastructure or surface features	
	E-scooter riders (n = 99)	Cyclists (n = 337)	E-scooter riders (n = 15)	Cyclists (n = 170)	E-scooter riders (n = 40)	Cyclists (n = 66)
Arrived by ambulance	39 (39.4)*	176 (52.2)	6 (40.0)	97 (57.1)	13 (32.5)*	34 (51.5)
Injury Severity Score (ISS) ^a	2.6 $\pm$ 2.0	3.4 $\pm$ 5.1	1.6 $\pm$ 1.2	2.5 $\pm$ 2.2	2.9 $\pm$ 2.0	3.6 $\pm$ 3.4
Splinting	47 (47.5)*	109 (32.3)	3 (20.0)	46 (27.1)	21 (52.5)	27 (40.9)
Fracture or dislocation reduction	12 (12.1)*	15 (4.5)	0	9 (5.3)	6 (15.0)*	2 (3.0)
Proximal upper extremity injury	9 (9.1)*	69 (20.5)	1 (6.7)	28 (16.5)	1 (2.5)*	18 (27.3)
Distal upper extremity injury	27 (27.3)	74 (22.0)	2 (13.3)	26 (15.3)	13 (32.5)	18 (27.3)
Distal lower extremity injury	13 (13.1)*	10 (3.0)	1 (6.7)	3 (1.8)	7 (17.5)*	3 (4.6)
Head or neck injury	7 (7.1)	25 (7.4)	2 (13.3)	8 (4.7)	2 (5.0)	5 (7.6)
Chest, abdomen, or spine injury	3 (3.0)*	47 (14.0)	1 (6.7)	27 (15.9)	2 (5.0)	8 (12.1)
External injury	54 (54.6)*	248 (73.6)	11 (73.3)	132 (77.7)	17 (42.5)*	44 (66.7)

*P < 0.05 for comparison between e-scooter riders and cyclists in logistic regression controlling for age and gender.

^aExcludes cases with unknown severity; <2% missing.

between e-scooter riders and cyclists in sustaining proximal upper extremity and distal lower extremity injuries were exaggerated in incidents involving surface conditions or infrastructure, although patterns persisted to a lesser degree when incidents involved motor vehicles. About three-quarters of riders in both groups sustained external injuries in incidents involving motor vehicles, and when injuries were due to surface features or infrastructure, external injuries were reduced among e-scooter riders and were sustained less frequently than among cyclists.

Cyclists were most often traveling on the road (104 [61.1%]) followed by in a bike lane (42 [24.7%]) prior to incidents involving vehicles, while e-scooter riders were most often on the sidewalk (7 [46.7%]) followed by the road (5 [33.3%]). Similarly, in incidents due to surface features or infrastructure, cyclists were most often on the road (37 [56.1%]) followed by the sidewalk or a trail or other off-road location (13 [19.7%] each), and e-scooter riders were most often on the sidewalk (23 [57.5%]) followed by the road (11 [27.5%]).

Estimated ED visits per mile traveled

We estimate that 20.7 (95% CI, 17.3–24.7) e-scooter riders sought treatment in the ED during enrollment hours in March to November per million miles traveled citywide, compared with 5.5 cyclists (95% CI, 5.0–6.0) (Table 5). More cyclists presented to the ED per month than e-scooter riders (20 cyclists compared with 15 e-scooter riders screened monthly), but estimated miles traveled were higher among cyclists. Higher ED visit rates per mile traveled for e-scooter riders persisted in sensitivity analyses where we used

different assumptions to estimate total cycling trips from commuting trips and restricted the sample to riders whose trips occurred entirely in Washington, DC (Table A5, online supplement).

Discussion

E-scooters and bicycles are both popular forms of micromobility, but the characteristics of riders injured on them and the ways in which they become injured differ substantially. We found that in Washington, DC, injured cyclists tend to be regular riders biking for utilitarian reasons, while injured e-scooter riders are more often infrequent users who are riding for social reasons. Cyclists are more frequently injured in the road in incidents involving motor vehicles, and e-scooter riders sustain more injuries on sidewalks and due to uneven surface features. Countermeasures to prevent cycling injuries implemented by cities primarily focus on providing facilities for cyclists to ride where they are separated from motor vehicles (DiGioia et al. 2017). Because the circumstances leading to e-scooter riders' injuries are so different, treatments such as bike lanes and other vehicle-separated facilities have the potential to prevent a smaller proportion of their injuries. E-scooters have a harder time handling uneven pavement and other obstacles because of their smaller tires, and their riders may especially benefit from better maintenance of the surfaces they use the most.

Cyclists in our sample were primarily male, similar to other literature on injured cyclists (National Highway Traffic Safety Administration 2020), while e-scooter riders were gender

Table 5. Estimated emergency room visits per mile traveled in Washington, DC, 2015–2017 (cyclists) or 2019 (e-scooter riders).

Type of rider	No. screened		Estimated no. trips		Mean trip length (miles)	Estimated miles traveled		ED visits per million miles traveled (95% CI)	RR (95% CI)
	Total	Per month	Total	Per month		Total	Per month		
E-scooter	121	15	5,573,966	669,757	1.05	5,852,664	703,245	20.7 (17.3, 24.7)	3.76 (3.08, 4.59)
Cyclist	483	20	21,428,372	904,753	4.10	87,856,326	3,709,489	5.5 (5.0, 6.0)	

CI, confidence interval; ED, emergency department; RR, relative risk.

balanced. E-scooter riders were also less likely to be ages 30–49 but more likely to be 50 and older compared with cyclists. These demographic differences could reflect in part differences in trip purposes or in the comparative ease of riding a bicycle and e-scooter. Cyclists were more often commuting, which is consistent with a greater number of them being of prime working ages. E-scooters require less physical exertion to ride than bicycles, and the age differences seen here mirror how a larger proportion of overall and injured e-bike riders are older adults than conventional cyclists (Ling et al. 2017; Otte and Facius 2020). Surveys of e-scooter riders in cities outside of Washington, DC, have reported younger ages and gender imbalance among users (Sanders et al. 2020), so it is unclear if our demographic results indicate greater injury risk among some cohorts or a different user base in the city.

We estimated that e-scooter riders are injured more often per mile that they travel, which is consistent with national trends (Watson et al. 2020) and reports observing more e-scooter riders than cyclists presenting to EDs during equivalent time periods (Trivedi et al. 2019; Puzio et al. 2020). It remains to be seen if this gap in injury rates will persist as e-scooter riders gain more experience and e-scooter sharing programs mature; for other modes, such as cycling (Poulos et al. 2015), inexperience increases crash risk. Over a third of injured e-scooter riders in our study were using a scooter for the first time. However, if e-scooter ridership growth persists, there will continue to be new, inexperienced riders well into the future.

Disparities in the types of injuries sustained likely reflect dissimilar injury mechanisms. Cyclists more often had injuries to the chest, abdomen, and spine, including some major intra-abdominal or intrathoracic injuries. These injuries may result from higher speeds among cyclists than e-scooters or handlebar position (Thompson and Rivara 2001). Greater proximal upper extremity injuries to cyclists could reflect injuries sustained when thrown from bicycles (Nelson and McKenzie 2011) and more lower distal extremity injuries to e-scooter riders could result from landing on their feet from a fall or using their feet to attempt to stop a fall. These injury patterns were heightened in incidents due to surface features or infrastructure, which were mostly falls.

A strength of this study is that we learned through interviews about injury circumstances and could confirm with patients that they were riding e-scooters, which can be challenging to identify retrospectively (English et al. 2020). This also introduced the limitation that riders who refused to be interviewed or presented after hours were excluded. Some riders excluded because they were too injured to communicate may have had more severe injuries and thus biased our sample toward lower injury severity; the proportion of serious injuries in our sample is consistent with some other emergency department and crash investigation studies of cyclists and e-scooter riders but lower than others (e.g., Davidson 2005; English et al. 2020; Otte and Facius 2020; Puzio et al. 2020).

Similar proportions of screened riders in each group were excluded for this reason (15 cyclists equaling 2.7% of cyclists screened, 3 e-scooter riders equaling 2.0% of e-scooter users screened), but it is possible that excluding these riders masked differences in injury severity and types between the groups.

E-scooter riders were given an incentive to participate and cyclists were not. This resulted in a greater proportion of screened e-scooter riders agreeing to be interviewed, and may also have concealed a difference in injury severity between the groups (i.e., higher injury severity among cyclists compared with e-scooter riders) if the incentive encouraged e-scooter riders with greater injury to participate. Seven cyclists who were trick riding or racing were excluded, which was not an exclusion criterion for e-scooter riders, and it is unclear how this difference affected injury outcomes. With the low top speed of e-scooters, we assume that racing was not common among e-scooter riders.

Cyclists were injured in different years than e-scooter riders because this study used cyclist data that had been previously collected for a different purpose; similarly, the sample of e-scooter riders was smaller than of cyclists, which limited statistical power. Injury circumstances can vary over time, and it is unknown how the distribution of scenarios leading to injury for cyclists may have differed in 2019. It is additionally unknown how accurately riders recalled the circumstances of their injury. In-depth crash investigations that include a more robust review of the scene and witnesses are needed to elucidate injury mechanisms; such investigations have not yet been conducted for e-scooter injuries.

Data from a single ED have limited generalizability. Washington, DC, requires that shared e-scooters be speed limited at 10 mph (16 km/h) citywide, which is unique among programs in major U.S. cities. Our estimate of ED visits per mile traveled include riders presenting to a single ED out of the five that serve adults in Washington, DC, but mileage represented travel citywide. Thus, the rates we report are smaller than the actual rate of riders treated in EDs throughout Washington, DC, and are lower than injury incident rates per mile traveled reported from comprehensive citywide surveillance of e-scooter riders treated in EDs in Austin, TX (English et al. 2020). The difference in rates between e-scooter riders and cyclists may be over- or underestimated if a larger proportion of use for one mode occurred more often near George Washington University Hospital relative to exposure patterns citywide.

Our values for cyclist miles traveled were extrapolated from commuting trip data and were a less accurate measure than what was available for e-scooter riders. However, ED presentation rates were consistently higher for e-scooter riders than cyclists when we varied the assumptions used to calculate them. Because mean trip lengths among cyclists were nearly four times as long as the average e-scooter trip, e-scooter riders would need to complete nearly three times as many trips as cyclists to result in equivalent ED presentation rates.

In summary, e-scooter riders in Washington, DC, are involved in more incidents leading to ED presentation per mile traveled than cyclists. Rider demographic characteristics, circumstances leading to injury, and injury types differ between these rider groups. Our findings suggest that offering well-maintained surfaces for e-scooters to ride on could decrease their injuries. Although we did not see differences in head injuries between cyclists and e-scooter riders, the fact that few e-scooter riders were helmeted suggests that promoting helmet use could also be beneficial to this group as helmets are well-established to reduce head injury risk for cyclists (Thompson et al. 1999). While some solutions to prevent cycling and e-scooter injuries are similar, such as the importance of helmets, the differences in how they are injured suggest that cyclists serve as an imperfect benchmark of what to expect in terms of e-scooter rider safety. Decisions regarding countermeasures to implement to make e-scooter riding safer should not be made solely based on what has worked for other, more established forms of micromobility such as cycling.

Acknowledgments

The authors would like to thank the research assistants who enrolled and interviewed the participants and the District Department of Transportation for providing e-scooter exposure data.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by the Insurance Institute for Highway Safety.

ORCID

Jessica B. Cicchino  <http://orcid.org/0000-0003-4337-518X>

References

- Badeau A, Carman C, Newman M, Steenblik J, Carlson M, Madsen T. 2019. Emergency department visits for electric scooter-related injuries after introduction of an urban rental program. *Am J Emerg Med*. 37(8):1531–1533. doi:10.1016/j.ajem.2019.05.003
- Cicchino JB, McCarthy ML, Newgard CD, Wall SP, DiMaggio CJ, Kulie PE, Arnold BN, Zuby DS. 2020. Not all protected bike lanes are the same: infrastructure and risk of cyclist collisions and falls leading to emergency department visits in three U.S. cities. *Accid Anal Prev*. 141:105490. doi:10.1016/j.aap.2020.105490
- Davidson JA. 2005. Epidemiology and outcome of bicycle injuries presenting to an emergency department in the United Kingdom. *Eur J Emerg Med*. 12(1):24–29. doi:10.1097/00063110-200502000-00007
- DiGioia J, Watkins KE, Xu Y, Rodgers M, Guensler R. 2017. Safety impacts of bicycle infrastructure: a critical review. *J Safety Res*. 61: 105–119. doi:10.1016/j.jsr.2017.02.015
- DiMaggio CJ, Bukur M, Wall SP, Frangos SG, Wen AY. 2020. Injuries associated with electric-powered bikes and scooters: analysis of US consumer product data. *Inj Prev*. 26(6):524–528. doi:10.1136/injury-prev-2019-043418
- English KC, Allen JR, Rix K, Zane DF, Ziebell CM, Brown CVR, Brown LH. 2020. The characteristics of dockless electric rental scooter-related injuries in a large U.S. city. *Traffic Inj Prev*. 21(7): 476–481. doi:10.1080/15389588.2020.1804059
- Gennarelli TA, Wodzin E. 2008. Abbreviated Injury Scale 2005: Update 2008. Des Plaines, IL: Association for the Advancement of Automotive Medicine;
- Ling Z, Cherry CR, MacArthur JH, Weinert JX. 2017. Differences of cycling experiences and perceptions between e-bike and bicycle users in the United States. *Sustainability*. 9(9):1662. doi:10.3390/su909
- Namiri NK, Lui H, Tangney T, Allen IE, Cohen AJ, Breyer BN. 2020. Electric scooter injuries and hospital admissions in the United States, 2014–2018. *JAMA Surg*. 155(4):357–359. doi:10.1001/jama-surg.2019.5423
- National Association of City Transportation Officials. 2020. Shared micromobility in the U.S.: 2019. <https://nacto.org/wp-content/uploads/2020/08/2020bikesharesnapshot.pdf>. Accessed September, 2020.
- National Highway Traffic Safety Administration. 2020. Traffic safety facts: bicyclists and other cyclists. <https://crashstats.nhtsa.dot.gov/Api/Public/ViewPublication/812884>. Accessed January, 2021.
- Nelson NG, McKenzie LB. 2011. Mountain biking-related injuries treated in emergency departments in the United States, 1994–2007. *Am J Sports Med*. 39(2):404–409. doi:10.1177/0363546510383478
- Otte D, Facius T. 2020. Accident typology comparisons between pedelecs and conventional bicycles. *J Transp Saf Secur*. 12(1):116–135. doi:10.1080/19439962.2019.1662530
- Poulos RG, Hatfield J, Rissel C, Flack LK, Murphy S, Grzebieta R, McIntosh AS. 2015. An exposure based study of crash and injury rates in a cohort of transport and recreational cyclists in New South Wales, Australia. *Accid Anal Prev*. 78:29–38. doi:10.1016/j.aap.2015.02.009
- Puzio TJ, Murphy PB, Gazzetta J, Dineen HA, Savage SA, Streib EW, Zarzaur BL. 2020. The electric scooter: a surging new mode of transportation that comes with risk to riders. *Traffic Inj Prev*. 21(2): 175–178. doi:10.1080/15389588.2019.1709176
- Sanders RL, Branion-Calles M, Nelson TA. 2020. To scoot or not to scoot: findings from a recent survey about the benefits and barriers of using E-scooters for riders and non-riders. *Transport Res A Pol*. 139:217–227. doi:10.1016/j.tra.2020.07.009
- Thompson DC, Rivara F, Thompson R. 1999. Helmets for preventing head and facial injuries in bicyclists. *Cochrane Database Syst Rev*. (4), 1–28. doi:10.1002/14651858.CD001855
- Thompson MJ, Rivara FP. 2001. Bicycle-related injuries. *Am Fam Physician*. 63(10):2007–2015.
- Trivedi TK, Liu C, Antonio ALM, Wheaton N, Kreger V, Yap A, Schriger D, Elmore JG. 2019. Injuries associated with standing electric scooter use. *JAMA Netw Open*. 2(1):e187381 doi:10.1001/jama-networkopen.2018.7381
- United States Census Bureau. American Community Survey. 2015, 2016, 2017. Commuting characteristics by sex, 1-year estimates subject table. Accessed April, 2020. <https://data.census.gov/cedsci/>.
- Watson HN, Garman CM, Wishart J, Zimmerman J. 2020. Patient demographics and injury characteristics of ER visits related to powered-scooters. *SAE Technical Paper* 2020-01-0933. doi:10.4271/2020-01-0933
- Zhang J, Yu KF. 1998. What's the relative risk? A method of correcting the odds ratio in cohort studies of common outcomes. *JAMA*. 280(19):1690–1691. doi:10.1001/jama.280.19.1690

APPENDIX

SUPPLEMENTAL METHODS: ED visit rate calculations

Number of ED visits

The number of emergency department (ED) visits that we used in rates per mile traveled included the adult cyclists (Figure A2) and e-scooter riders (Figure A3) who were screened during March to November and injured in the hospital's catchment area within the past week. Screened patients who were eligible and refused participation or withdrew, or were ineligible for enrollment because they were not English-speaking, too injured to communicate, unable to understand consent, left the ED without being seen by a physician, riding with more than one person on a bicycle or e-scooter, ineligible for a reason not recorded, or bicyclists trick riding or racing were included in the count of ED visits. Potential participants who were missed prior to screening, not injured while riding a bicycle or e-scooter, under age 18 years, injured outside of the hospital's catchment area, injured more than one week before their ED visit, or previously enrolled in the study were excluded. This process resulted in 483 cyclists and 121 e-scooter riders.

E-scooter exposure

Shared e-scooter operators in Washington, DC, submitted monthly data on the length and date of each trip taken to the District Department of Transportation (DDOT) and monthly summaries of the total number of trips. We extracted the number of trips during March 23 to November 30, 2019 from the data on individual trips and used counts from the monthly summary data for operators from whom trip-level data were unavailable. We adjusted the 4,737,871 trips reported by DDOT during this period to reflect that 85% of e-scooter riders in our sample reported riding shared e-scooters (Table A1), resulting in an adjusted total of 5,573,966 trips. The adjusted number of trips was multiplied by the average trip length among enrolled e-scooter participants of 1.05 miles (Table A1) to calculate miles traveled.

Bicycle exposure

There is no comprehensive measure of cycling exposure that exists for Washington, DC, and so we extrapolated cycling miles traveled from available sources. Our method of estimating cycling trips

was derived loosely from Pucher and Buehler (2016) and the New York City Department of Transportation (2019). We obtained the number of people who commute by bicycle in Washington, D.C., from the 2015, 2016, and 2017 American Community Survey, which is a yearly survey administered by the United States Census Bureau (2020) with an annual sample size of over 4,500 in Washington, DC, and with sampling weights used to compute population-level counts. We assumed that the average of 16,663 bicycle commuters (14,718 commuters in 2015, 16,647 in 2016, 18,624 in 2017; United States Census Bureau 2015, 2016, 2017) each made 339 commute trips per year, which was the mean number of annual commute trips per U.S. worker reported in the 2017 National Household Travel Survey (McGuckin & Fucci 2018), a nationally-representative survey of travel behavior that sampled 130,000 households. Based on these figures, we estimated that cyclists in Washington, DC, completed 5,648,757 commuting trips annually.

We assumed that the share of cycling trips that were made for commuting in Washington, DC, was equal to the share among enrolled study cyclists (52%; Table A1), and extrapolated that cyclists completed 10,862,994 total trips annually. There were 720 days when bicycle data were collected during the months of March to November, which resulted in an estimate of 21,428,372 total trips during these months of the study period. Trips were multiplied by the average cyclist trip length among enrolled cyclists of 4.10 miles (Table A1).

Sensitivity Analyses

We also calculated injury rates under different trip length and bicycling commute share assumptions (Table A5). Our exposure data were drawn from Washington, DC, but study cyclists could have been injured in or part of their trip could have occurred in the greater metropolitan area. Average cycling trip lengths were shorter for cyclists whose entire trips occurred in Washington, DC, which would result in fewer total miles traveled. We multiplied the number of ED visits used in the primary incident rates by the proportion of enrolled riders with trips completely entirely in the District (Table A1; 94% of e-scooter riders and 73% of cyclists) to estimate that there were 114 e-scooter riders and 353 cyclists who presented to the ED where all exposure occurred in Washington, DC.

Among the enrolled riders whose trips occurred entirely within Washington, DC, mean trip length was 1.05 (SD 1.15) miles for e-scooter riders and 2.33 (SD 3.80) miles for cyclists. We additionally extracted trip lengths for all shared e-scooter trips completed in Washington, DC, from DDOT trip data; data were cleaned to remove implausibly long values of 100 miles or greater. Mean e-scooter trip length from this source was 0.93 (SD 1.12) miles. We used these trip lengths to compute ED visit rates per mile traveled for riders estimated to have only traveled in Washington, DC, prior to sustaining their injuries.

Our method of extrapolating total cycling trips from commuting trips was more conservative than prior studies, which assumed that bicycle commuters took two trips daily and that commuting trips made up 20% of all cycling trips (New York City Department of Transportation 2019; Pucher & Buehler 2016). Table A5 shows that the total estimated number of cycling trips would be larger, and thus cycling ED visit rates per mile traveled would be lower, if we assumed that 20% of trips were for commuting.

SUPPLEMENTAL REFERENCES

McGuckin N, Fucci A. Summary of travel trends: 2017 National Household Travel Survey. Washington, DC: Federal Highway Administration; 2018.

New York City Department of Transportation. Cycling in the city: cycling trends in NYC, May 2019. 2019; <https://www1.nyc.gov/html/dot/downloads/pdf/cycling-in-the-city.pdf>. Accessed June, 2020.

Pucher J, Buehler R. Safer cycling through improved infrastructure. *Am J Public Health*. 2016;106(12):2089-2091. doi: 10.2105/AJPH.2016.303507

United States Census Bureau. American Community Survey, Commuting characteristics by sex, 1-year estimates subject table. 2015, 2016, 2017. Accessed April, 2020.

United States Census Bureau. American Community Survey: Methodology. 2020; <https://www.census.gov/programs-surveys/acs/methodology.html>. Accessed June, 2020.

SUPPLEMENTAL TABLES

Table A1. Demographic and trip characteristics of cyclists (2015-2017) and e-scooter riders (2019) seeking treatment in a Washington, DC, emergency department

Characteristic	No. (%) or Mean ± SD		P Value ^a
	E-scooter riders (n=99)	Cyclists (n=337)	
Demographic characteristics			
Age (years)			
18-29	33 (33.3)	98 (29.1)	.01
30-49	32 (32.3)	163 (48.4)	
50+	34 (34.3)	76 (22.6)	
Female	45 (45.5)	98 (29.1)	.002
Race/ethnicity ^b			
White, non-Hispanic	61 (62.9)	232 (70.1)	.56
Black, non-Hispanic	13 (13.4)	45 (13.6)	
Hispanic	12 (12.4)	26 (7.9)	
Asian/Pacific Islander	5 (5.2)	13 (3.9)	
Other	6 (6.2)	15 (4.5)	
College degree or higher ^b	70 (70.7)	274 (81.6)	.02
Trip characteristics			
Wore helmet ^b	2 (2.0)	223 (66.4)	<.001
Rode shared or rental e-scooter/bike ^b	83 (84.7)	23 (6.9)	<.001
Mean trip length (miles) ^c	1.05 ± 1.13	4.10 ± 6.25	<.001
Entire trip in Washington, DC ^d	81 (94.2)	243 (72.5)	<.001
Weekend	32 (32.3)	58 (17.2)	.001
Time of day ^b			
7 AM-2:59 PM	56 (57.1)	232 (68.8)	.006
3 PM-10:59 PM	30 (30.6)	91 (27.0)	
11 PM-6:59 AM	12 (12.2)	14 (4.2)	
Riding frequency ^{b,e}			
First time	35 (35.7)	NA	NA
Less than weekly	17 (17.4)	18 (5.4)	
At least once a week, but not most days	15 (15.3)	53 (15.8)	
Most days	31 (31.6)	265 (78.9)	
Trip purpose ^b			
Social reasons (e.g., movies, visit friends)	35 (35.4)	36 (10.7)	<.001
Commute to/from work	25 (25.3)	175 (52.1)	
Recreation	19 (19.2)	69 (20.5)	
Personal business (e.g., errands)	18 (18.2)	35 (10.4)	
To/from school	0	6 (1.8)	
Other	2 (2.0)	15 (4.5)	
Intersection ^b	35 (34.7)	114 (33.8)	.87
Location ^b			
Sidewalk	55 (56.1)	51 (15.1)	<.001
Road	24 (24.5)	171 (50.7)	
Off-road/multiuse trail	9 (9.2)	51 (15.1)	
Bike lane	8 (8.2)	60 (17.8)	
Alley, driveway, parking lot	2 (2.0)	4 (1.2)	

Table A1. Demographic and trip characteristics of cyclists (2015-2017) and e-scooter riders (2019) seeking treatment in a Washington, DC, emergency department (continued)

Characteristic	No. (%) or Mean $\pm$ SD		P Value ^a
	E-scooter riders (n=99)	Cyclists (n=337)	
Injury circumstances ^f			
Moving vehicle ^g	13 (13.1)	127 (37.7)	<.001
Parked or stopped vehicle	2 (2.0)	43 (12.8)	
Pedestrian	11 (11.1)	17 (5.0)	
Cyclist or scooter ^h	3 (3.0)	22 (6.5)	
Surface feature (e.g., potholes, uneven pavement)	24 (24.2)	48 (14.2)	
Infrastructure (e.g., curb, pole, fence)	16 (16.2)	18 (5.3)	
Other fall	26 (26.3)	59 (17.5)	
Other circumstance	4 (4.0)	3 (0.9)	

Abbreviation: NA, not applicable.

^aDetermined through χ^2 test or proportions and *t* test for means.

^bProportions exclude cases with missing data; <2% missing

^cData on trip length were missing for 12 e-scooter riders (12.1%).

^dData on trip location were missing for 13 e-scooter riders (13.2%) and 2 (0.6%) cyclists.

^eResponses are not compared by user type because the question differed. E-scooter riders were asked how often they rode during the past month; cyclists were asked how often they ride during the season when they ride the most.

Cyclists did not have a response option of “first time”.

^fUnless specifically labeled as a fall, circumstances included crashes with and falls to avoid listed element
^g10 (10.1%) e-scooter riders and 111 cyclists (32.9%) crashed with moving vehicles.

^h1 e-scooter rider collided with another scooter. Other crashes and falls in this category involved collisions with or falls to avoid cyclists.

Table A2. Evaluation, treatment, and injury characteristics of cyclists (2015-2017) and e-scooter riders (2019) seeking treatment in a Washington, DC, emergency department

Characteristic	No. (%) or Mean $\pm$ SD		RR (95% CI) ^a
	E-scooter riders (n=99)	Cyclists (n=337)	
Arrived by ambulance	39 (39.4)	176 (52.2)	0.62 (0.41, 0.91)
Imaging performed in ED			
Head or neck	29 (29.3)	81 (24.0)	1.20 (0.80, 1.69)
Chest, abdomen, or spine	17 (17.2)	138 (41.0)	0.35 (0.20, 0.59)
Extremity	66 (66.7)	226 (67.1)	0.96 (0.64, 1.37)
FAST	4 (4.0)	16 (4.8)	1.01 (0.38, 2.14)
Procedures performed in ED			
Wound care	36 (36.4)	154 (45.7)	0.74 (0.49, 1.08)
Laceration repair	24 (24.2)	81 (24.0)	1.02 (0.66, 1.49)
Splinting	47 (47.5)	109 (32.3)	1.69 (1.23, 2.20)
Fracture or dislocation reduction	12 (12.1)	15 (4.5)	2.06 (1.22, 2.94)
Admitted or transferred	9 (9.1)	27 (8.0)	1.14 (0.59, 1.94)
Procedures performed in hospital			
Orthopedic	5 (5.1)	15 (4.5)	1.24 (0.52, 2.20)
Neurosurgical	0	1 (0.3)	NA
Other operative procedure	1 (1.0)	4 (1.2)	0.87 (0.11, 3.10)
ISS ^b	2.6 $\pm$ 2.0	3.4 $\pm$ 5.1	0.94 (0.88, 1.08)
Body region injured ^c			
Head or neck ^d	7 (7.1)	25 (7.4)	0.96 (0.45, 1.79)
AIS 2+ head injury	4 (4.0)	5 (1.5)	2.01 (0.77, 3.38)
AIS 2+ concussion (with LOC)	4 (4.0)	2 (0.6)	3.03 (1.20, 4.09)
Skull fracture	2 (2.0)	0	NA
Intracranial hemorrhage	1 (1.0)	3 (0.9)	0.93 (0.11, 3.24)
Face	9 (9.1)	20 (5.9)	1.34 (0.70, 2.23)
Chest, abdomen, or spine	3 (3.0)	47 (14.0)	0.24 (0.07, 0.70)
Extremity	49 (49.5)	153 (45.4)	1.21 (0.85, 1.66)
Upper extremity	34 (34.3)	137 (40.7)	0.86 (0.58, 1.24)
Upper extremity, proximal ^e	9 (9.1)	69 (20.5)	0.49 (0.24, 0.92)
Upper extremity, distal ^f	27 (27.3)	74 (22.0)	1.28 (0.86, 1.81)
Lower extremity	16 (16.2)	17 (5.0)	2.28 (1.49, 3.05)
Lower extremity, proximal ^g	3 (3.0)	7 (2.1)	1.13 (0.35, 2.57)
Lower extremity, distal ^h	13 (13.1)	10 (3.0)	2.76 (1.79, 3.54)
External	54 (54.6)	248 (73.6)	0.47 (0.30, 0.71)

Abbreviations: AIS, abbreviated injury scale; CI, confidence interval; ED, emergency department; FAST, focused assessment with sonography for trauma; ISS, Injury Severity Score; LOC, loss of consciousness; NA, not applicable; RR, relative risk.

^aAdjusted for age and gender.

^bProportions exclude cases with unknown severity; <2% missing.

^cExternal injuries include lacerations, contusions, and abrasions, which are assigned to the external body region when calculating the Injury Severity Score (ISS). Injuries reported to other body regions exclude external injuries and otherwise are categorized based on the AIS rather than the ISS body regions.

^d1 cyclist experienced a neck injury. Other injuries in this category are head injuries.

^eProximal upper extremity includes clavicle, scapula, shoulder, humerus.

^fDistal upper extremity includes elbow, forearm, wrist, hand, finger.

^gProximal lower extremity includes pelvis, hip, femur.

^hDistal lower extremity includes knee, tibia/fibula, ankle, foot.

Table A3. Evaluation, treatment, and injury characteristics of cyclists (2015-2017) and e-scooter riders (2019) seeking treatment in a Washington, DC, emergency department due to incidents involving motor vehicles

Characteristic	No. (%)		RR (95% CI) ^a
	E-scooter riders (n=15)	Cyclists (n=170)	
Arrived by ambulance	6 (40.0)	97 (57.1)	0.60 (0.20, 1.66)
Imaging performed in ED			
Head or neck	5 (33.3)	39 (22.9)	1.59 (0.54, 3.98)
Chest, abdomen, or spine	2 (13.3)	75 (44.1)	0.22 (0.04, 0.96)
Extremity	9 (60.0)	109 (64.2)	0.75 (0.25, 2.03)
FAST	0	9 (5.3)	NA
Procedures performed in ED			
Wound care	7 (46.7)	77 (45.3)	1.22 (0.43, 3.05)
Laceration repair	3 (20.0)	31 (18.2)	1.50 (0.41, 4.44)
Splinting	3 (20.0)	46 (27.1)	0.85 (0.23, 2.73)
Fracture or dislocation reduction	0	9 (5.3)	NA
Admitted or transferred	0	10 (5.9)	NA
Procedures performed in hospital			
Orthopedic	0	4 (2.4)	NA
Neurosurgical	0	1 (0.6)	NA
Other operative procedure	0	1 (0.6)	NA
ISS ^b	1.6 ± 1.2	2.5 ± 2.2	0.81 (0.55, 1.16)
Body region injured ^c			
Head or neck ^d	2 (13.3)	8 (4.7)	1.88 (0.39, 6.16)
AIS 2+ head injury	0	0	NA
AIS 2+ concussion (with LOC)	0	0	NA
Skull fracture	0	0	NA
Intracranial hemorrhage	0	0	NA
Face	0	6 (3.5)	NA
Chest, abdomen, or spine	1 (6.7)	27 (15.9)	0.38 (0.05, 2.54)
Extremity	4 (26.7)	58 (34.1)	0.86 (0.26, 2.53)
Upper extremity	3 (20.0)	52 (30.6)	0.77 (0.21, 2.52)
Upper extremity, proximal ^e	1 (6.7)	28 (16.5)	0.46 (0.06, 3.01)
Upper extremity, distal ^f	2 (13.3)	26 (15.3)	1.11 (0.24, 4.07)
Lower extremity	1 (6.7)	6 (3.5)	1.33 (0.15, 6.75)
Lower extremity, proximal ^g	0	3 (1.8)	NA
Lower extremity, distal ^h	1 (6.7)	3 (1.8)	1.93 (0.19, 8.53)
External	11 (73.3)	132 (77.7)	0.80 (0.24, 2.41)

Abbreviations: AIS, abbreviated injury scale; CI, confidence interval; ED, emergency department; FAST, focused assessment with sonography for trauma; LOC, loss of consciousness; NA, not applicable; RR, relative risk.

^aAdjusted for age and gender.

^bProportions exclude cases with unknown severity; <3% missing.

^cExternal injuries include lacerations, contusions, and abrasions, which are assigned to the external body region when calculating the Injury Severity Score (ISS). Injuries reported to other body regions exclude external injuries and otherwise are categorized based on the AIS rather than the ISS body regions.

^d1 cyclist experienced a neck injury. Other injuries in this category are head injuries.

^eProximal upper extremity includes clavicle, scapula, shoulder, humerus.

^fDistal upper extremity includes elbow, forearm, wrist, hand, finger.

^gProximal lower extremity includes pelvis, hip, femur.

^hDistal lower extremity includes knee, tibia/fibula, ankle, foot.

Table A4. Evaluation, treatment, and injury characteristics of cyclists (2015-2017) and e-scooter riders (2019) seeking treatment in a Washington, DC, emergency department due to incidents involving surface conditions or infrastructure

Characteristic	No. (%)		RR (95% CI) ^a
	E-scooter riders (n=40)	Cyclists (n=66)	
Arrived by ambulance	13 (32.5)	34 (51.5)	0.50 (0.23, 0.94)
Imaging performed in ED			
Head or neck	10 (25.0)	14 (21.2)	1.13 (0.59, 1.75)
Chest, abdomen, or spine	6 (15.0)	25 (37.9)	0.39 (0.15, 0.87)
Extremity	27 (67.5)	44 (66.7)	0.96 (0.51, 1.53)
FAST	2 (5.0)	1 (1.5)	1.94 (0.49, 2.57)
Procedures performed in ED			
Wound care	15 (37.5)	33 (50.0)	0.62 (0.31, 1.12)
Laceration repair	9 (22.5)	22 (33.3)	0.66 (0.30, 1.21)
Splinting	21 (52.5)	27 (40.9)	1.34 (0.83, 1.85)
Fracture or dislocation reduction	6 (15.0)	2 (3.0)	2.10 (1.10, 2.53)
Admitted or transferred	5 (12.5)	7 (10.6)	1.08 (0.52, 1.99)
Procedures performed in hospital			
Orthopedic	3 (7.5)	5 (7.6)	1.14 (0.37, 2.05)
Neurosurgical	0	0	NA
Other operative procedure	1 (2.5)	2 (3.0)	1.00 (0.13, 2.33)
ISS ^b	2.9 ± 2.0	3.6 ± 3.4	0.95 (0.85, 1.04)
Body region injured ^c			
Head ^d	2 (5.0)	5 (7.6)	0.84 (0.21, 1.91)
AIS 2+ head injury	2 (5.0)	1 (1.5)	1.88 (0.45, 2.56)
AIS 2+ concussion (with LOC)	2 (5.0)	0	NA
Skull fracture	1 (2.5)	0	NA
Intracranial hemorrhage	0	1 (1.5)	NA
Face	4 (10.0)	5 (7.6)	1.16 (0.42, 2.02)
Chest, abdomen, or spine	2 (5.0)	8 (12.1)	0.55 (0.13, 1.53)
Extremity	21 (52.5)	39 (59.1)	0.91 (0.50, 1.44)
Upper extremity	14 (35.0)	35 (53.0)	0.60 (0.30, 1.07)
Upper extremity, proximal ^e	1 (2.5)	18 (27.3)	0.12 (0.02, 0.72)
Upper extremity, distal ^f	13 (32.5)	18 (27.3)	1.10 (0.60, 1.68)
Lower extremity	7 (17.5)	4 (6.1)	1.93 (1.08, 2.42)
Lower extremity, proximal ^g	0	1 (1.5)	NA
Lower extremity, distal ^h	7 (17.5)	3 (4.6)	2.07 (1.20, 2.49)
External	17 (42.5)	44 (66.7)	0.45 (0.22, 0.86)

Abbreviations: AIS, abbreviated injury scale; CI, confidence interval; ED, emergency department; FAST, focused assessment with sonography for trauma; LOC, loss of consciousness; NA, not applicable; RR, relative risk.

^aAdjusted for age and gender.

^bProportions exclude cases with unknown severity; <2% missing.

^cExternal injuries include lacerations, contusions, and abrasions, which are assigned to the external body region when calculating the Injury Severity Score (ISS). Injuries reported to other body regions exclude external injuries and otherwise are categorized based on the AIS rather than the ISS body regions.

^dNo neck injuries were due to surface features or infrastructure.

^eProximal upper extremity includes clavicle, scapula, shoulder, humerus.

^fDistal upper extremity includes elbow, forearm, wrist, hand, finger.

^gProximal lower extremity includes pelvis, hip, femur.

^hDistal lower extremity includes knee, tibia/fibula, ankle, foot.

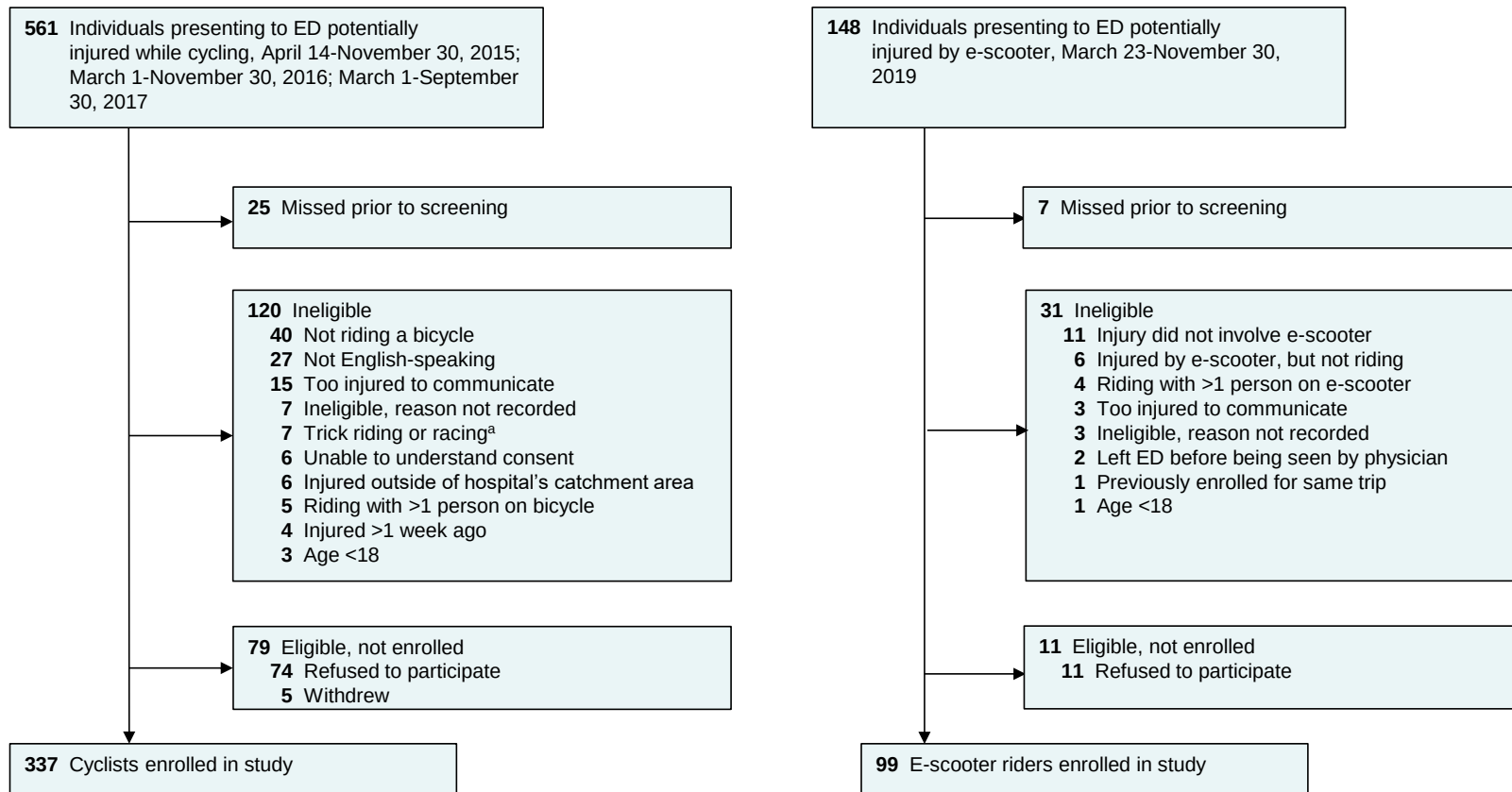
Table A5. Estimated emergency department visits per million miles traveled for e-scooter riders and cyclists under various assumptions, 2015-2017 (Cyclists) or 2019 (E-scooter Riders)

Type of rider	Type of trip	No. screened	% of trips that were commuting	Estimated no. trips	Mean trip length (miles)	Estimated miles traveled	ED visits per million miles traveled (95% CI)
E-scooter	All	121	NA	5,573,966	1.05	5,852,664	20.7 (17.3, 24.7)
	Entirely in DC	114			1.05	5,852,664	19.5 (16.2, 23.4)
					0.93	5,183,788	22.0 (18.3, 26.4)
Cyclist	All	483	52%	21,428,372	4.10	87,856,326	5.5 (5.0, 6.0)
	Entirely in DC	353			2.33	49,928,107	7.1 (6.4, 7.9)
	All	483	20%	55,713,768	4.10	228,426,447	2.1 (1.9, 2.3)
	Entirely in DC	353			2.33	129,813,079	2.7 (2.4, 3.0)

Abbreviations: CI, confidence interval; ED, emergency department.

SUPPLEMENTAL FIGURES

Figure A1. Identification of study sample from patients presenting to Washington, DC, emergency department during 2015-2017 (cyclists) or 2019 (e-scooter riders).

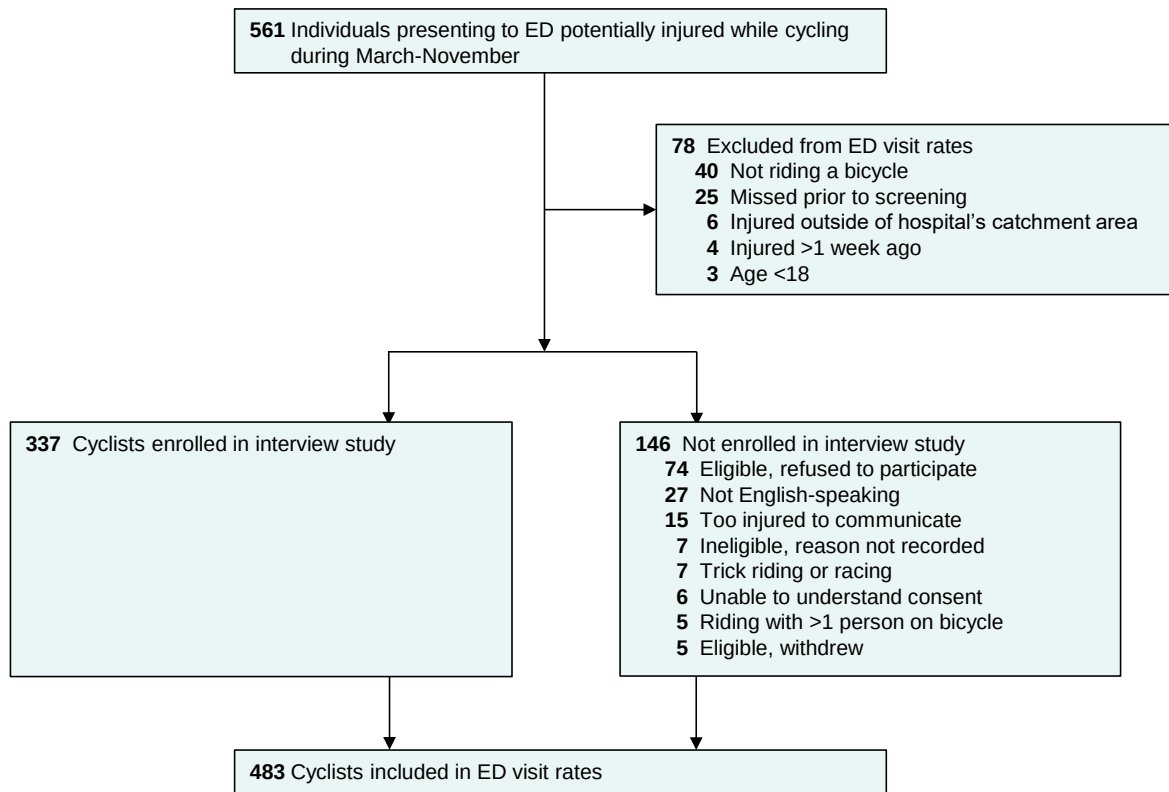


Abbreviations: ED, emergency department.

Note: Data were collected during hours that research assistants staffed the ED (8:00 AM-10:00 PM during 2015-2017, 9:00 AM-10:00 PM during 2019); patients presenting outside these hours were not captured.

^aTrick riding or racing was exclusion criterion for bicyclists only.

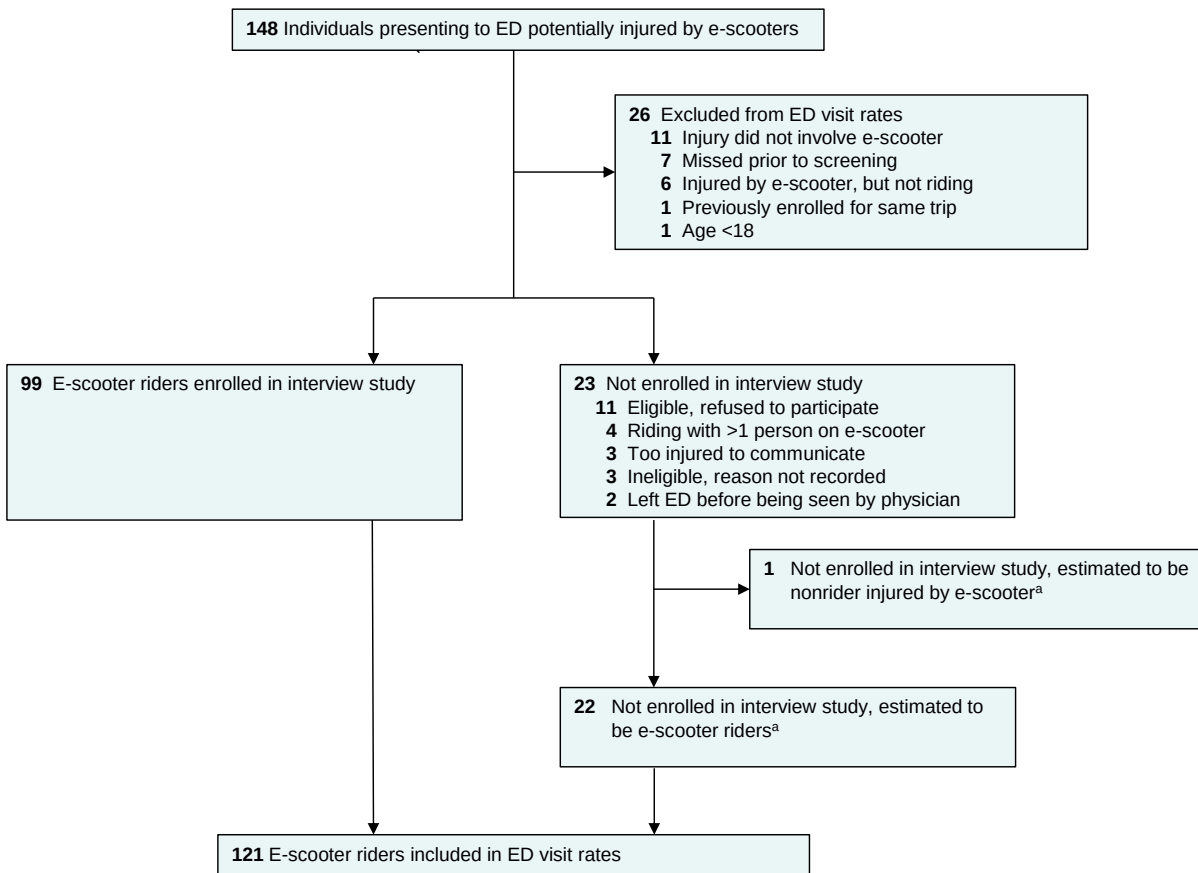
Figure A2. Identification of cyclist ED visits used in rates per mile traveled from patients presenting to Washington, DC, emergency department, 2015-2017



Abbreviations: ED, emergency department

Note: Data were collected during hours that research assistants staffed the ED (8:00 AM-10:00 PM); patients presenting outside these hours were not captured.

Figure A3. Identification of e-scooter rider ED visits used in rates per mile traveled from patients presenting to Washington, DC, emergency department, 2015-2017



Abbreviations: ED, emergency department

Note: Data were collected during hours that research assistants staffed the ED (9:00 AM-10:00 PM); patients presenting outside these hours were not captured.

^aA total of 6% (6 of 105) of individuals injured by e-scooters that we interviewed were nonriders. We assumed that 6% of individuals injured by e-scooters who were ineligible for the interview study (1 of 19) were nonriders.